

SYNTHETIC CONTROL

Generalized synthetic controls (gsynth)

- Method distinction:
 - Unifies original synthetic control with interactive fixed effects model, DiD is a special case.
 - “Generalized” because it extends original synthetic control to the case of multiple treated units and variable treatment periods.
 - The general steps are:
 - 1. Use untreated observations to estimate factors in a latent factor model: common factors that vary with time and influence outcomes. The number of factors can be chosen via cross-validation.
 - 2. Use treated units pre-treated data to fit their factor loadings (these vary by unit but not over time).
 - 3. Use the now fully fit factor model to predict control outcomes post-treatment for treated units
 - Because of the parametric setup, it’s more likely that a good synthetic control can’t be constructed here - important to validate **[this may be why confidence intervals are sometimes so large]**
 - Covariates can be included for learning the factor model, potentially improving precision
 - I don’t see why the matrix completion method wouldn’t work in theory for one treated unit, but [Yiqing Xu commented this on Github](#), and for sure it is not currently implemented for the package
- Standard error methods:
 - Parametric bootstrap is used for confidence intervals (non-parametric option, but the documentation recommends parametric when the number of treated units is small)
 - In each iteration, randomly select one control unit as if it were treated instead
 - Resample the rest of the control group with replacement
 - Apply GSC to the new sample, getting a vector of prediction errors e^P for the placebo treated unit
 - Apply GSC to original data, getting vector of prediction errors e for each control unit
 - For a number of iterations:
 - Start with the GSC original model predictions for control units, with “control” set that samples w/ replacement an added error from e and a “treated” set that samples w/ replacement an added error from e^P . This is a fully placebo sample - no real treatment effects.
 - Apply GSC method to S and get an ATT estimate, add this to the GSC original method estimate
 - Use these to create permutation distribution for p-values, CIs

Micro synthetic controls (microsynth)

- Method distinction:
 - Allows multiple outcomes to be used at once - assumes unobserved confounders are common to all outcomes.
 - But, our current use case is one outcome, one treated unit: this package just boils down to conventional synthetic control in that case, possibly with covariates used in the matching/alignment algo.
- Standard error methods:
 - They offer three methods: linearization, jackknife, and permutation. They say linearization is “best practice” in general, but not trustworthy with a single treated unit. Similarly, they advise not to use jackknife in this case ([see their JSS paper p. 23](#)).
 - They advise only to use p-values from the permutation approach, but not confidence intervals (package won't return confidence intervals from permutation in our case) ([p. 25](#))

[Augmented synthetic controls \(augsynth\)](#)

- Method distinction:
 - Uses an outcome model to estimate bias due to imperfect pre-treatment fit, de-biases the original synthetic control method.
 - Covariates can be included in both the original synthetic controls model and the outcome model for estimating bias.
 - Doubly robust - if either the synthetic control reweighting or the outcome correction model works well, low bias
- Standard error methods:
 - Conformal option (refer to [Chernozhukov et al. 2021](#) - I found this clearer than the aug synth paper)
 1. For a given sharp null hypothesis $\tau = \tau_0$, edit observation of treated unit $Y_{1T} - \tau_0$ as the “new” treatment data (we usually just consider $\tau_0 = 0$ so no change).
 2. Apply the augsynth method to the adjusted data set to get adjusted weights and treated unit residuals under “control” in post-treatment periods, then aggregate residuals for the test stat (bigger $\rightarrow$ rejection)
 3. Compute p-value: what proportion of times is the treated outcome residual from step 2 larger than residuals when we permute time periods?
 4. Compute confidence intervals: repeat this for a fine grid of τ_0 values, the set of τ_0 values in which $p > \alpha$ (where we cannot reject the sharp null) is our pointwise confidence interval.
 5. Practical notes: this only appears to be implemented for some of the bias correction methods, does not apply to: elastic net (EN), random forest (RF) or neural networks (seq2seq)
- Jackknife and [jackknife+](#)

- Mentioned in the [github for the package](#), but not in the paper. They note it requires additional assumptions but they give no theoretical results or explicit assumptions on how they perform in this setting for synthetic control.
- Jackknife+ is a variation on jackknife (leave-one-out predictions) where you just use the LOO prediction as the point estimate, not the estimate from the full data model. Based on the documentation, jackknife+ seems to be implemented to leave out one **time period**, while jackknife leaves out one **unit**.